

Utilization of Local Phasor Measurements for Interarea-Oscillation Damping with Utility-Scale PV Plant

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Abstract – This paper proposes a power oscillation damping (POD) control scheme utilizing a photovoltaic (PV) power plant, improving the power systems' dynamic response to interarea oscillations, particularly for high-renewable and low-inertia systems. To achieve this, two separate control loops, an active power modulation loop without curtailment and another loop with reactive power current injection, have been utilized in a two-area system. The feedback for these damping controllers is taken from Phasor Measurement Units (PMUs) located far from the controllers, making these signals highly susceptible to communication time delays, failures, and cyber-attacks that may compromise controller performance. This paper uses the local bus angle and bus voltage at the PV plant's point of interconnection (POI) as the feedback for each of these loops. We verified the effectiveness of the proposed control scheme with a three-phase short circuit fault in a two-area power system. We also compared the proposed method with the conventional technique to investigate the superiority of the proposed controller.

Keywords—microgrid, fault restoration, fault ride through, dynamic response, transient stability, PSCAD, photovoltaic (PV), PV penetration

I. INTRODUCTION

The increasing demand for electrical energy in the 21st century, with the limited fossil fuels and increasing environmental concerns of carbon dioxide and greenhouse gas emission, calls for fast and adaptive development in the area of renewable energy sources (RESs) [1]–[2]. Hence, the integration of RESs, especially solar photovoltaics (PVs), has increased significantly in the recent past in the modern power grid [3]. Globally, the installed capacity of solar PV plants has increased rapidly from 3.7 GW in 2004 to 509 GW in 2018 [4]. Solar PV is expected to grow the fastest from now to 2050 of all the RESs [5]. However, the rapid expansion of Inverter

Based Resources (IBRs) might affect the system stability depending upon the penetration level, location, and control schemes [6]–[8].

The intermittent nature of the RESs is one of the major concerns in power system studies. Moreover, the inertial and dynamic characteristics of IBRs are much different compared to the conventional synchronous machines-based power generations. Unlike conventional generators, PV plants do not possess the inherent capability to provide inertia to the system, making the grid system more vulnerable during large frequency excursions. IBRs have power electronic converters at the grid interface, which decouples the dynamic of PV plant from the AC grid system [9]–[10]. Moreover, IBRs operate on maximum power point tracking. Hence, it cannot provide increased power output when needed [11]. Therefore it is apparent that the PV plants inherently can neither provide the inertial response nor can participate in frequency regulation. Thus, technical issues regarding frequency regulation, voltage stability, and fault ride-through (FRT) have become an integral part of power systems with high penetrated PV plants [12]–[14]. It is imminent to shift the focus of grid-supporting control schemes for PV plants regarding grid disturbances. So, it has become imminent to shift the focus in auxiliary control schemes of PV plants to provide some promising solutions to its shortcomings.

A grid with less inertia is always prone to severe oscillation, even with a small load change on its terminal. These low-frequency electromechanical power oscillations (typically 0.1–2 Hz) are often damped out with reactive power of the FACTS devices by increasing the power transfer capabilities in the transmission line. The effectiveness of various FACTS devices, such as Static Var Compensators (SVC), Static Synchronous Compensators (STATCOM), Thyristor

Controlled Series Compensators (TCSC), and Convertible Static Compensator (CSC), in power oscillation damping, are verified [15]–[19]. Exploiting ESSs, e.g., battery energy storage (BESS), supercapacitor storage (SCESS), has also been a common practice in responding to system frequency variations. The ESSs with proposed control parameter tuning of active power output have successfully dampened low-frequency oscillations in [20] and [21]. However, additional equipment demands a capital cost of the grid-connected PV plants. Therefore, some studies modify the existing inverter control to minimize the burden from the other devices [22].

Enabling PV plants to participate on the power oscillation damping might be a cost-effective solution in this case. The power converters in PV plants consist of two decoupled parallel loops which independently can control the active and reactive power. In [23]–[24], a reactive current loop of the PV plant has been modified to provide efficient damping of power oscillation. However, they placed the PV plant in the middle of the transmission line in both cases, which is not a feasible situation in all cases. In [25], the PV plant is used to damp local power oscillation in the distribution lines, and questions have been raised about its use in inter-area oscillation damping. In some research work [26]–[27], an active power control loop has been utilized to damp inter-area oscillation and has shown promising and fast performances in damping. However, curtailment of the active power of the PV plants was common in all these cases. In [28]–[29], both active and reactive power loop has been used for power oscillation damping. Results have shown that this might be the fastest low-frequency oscillation damping method. However, in both cases, external signals from remote PMUs have been taken as feedback. A communication delay was prominent in both cases. This is where wide-area management (WAMS) comes into account. Various algorithms, such as quantum particle swarm optimization (QSP0) [30], adaptive neuro-fuzzy logic (ANFL) [31], etc. have been used to adjust this communication delay and make the controllers faster. Hence, a faster and more efficient solution would be to use local PMU signals as feedback to damp out the low-frequency oscillations.

This paper proposes a damping control scheme of a PV to secure the power system resilience for grid faults using local bus phase angle measurement. The POD is activated upon detecting system frequency oscillations. In this paper, three power oscillation damping (POD) control schemes are discussed and compared. The damping control schemes are 1) Reactive POD control, 2) Active POD control, 3) a combination of both Reactive and Active Power Damping control schemes. The proposed POD enables the PV plant to support the voltage stability for a grid fault and provide a fast system stabilization after the fault clearance. The model of the control scheme is presented in section II. Once the control scheme has been modeled, simulation testing is introduced in section III. The proposed control scheme improves the resiliency of the system and neglects the additional capital

investment cost as presented on the analysis are also in the same section. At last, this work discusses the further works of the proposed scheme for better implementation.

II. PROPOSED CONTROL SCHEME FOR IBR

A. PV System Model

The configuration of a PV system model used in this paper is depicted in Fig. 1. The PV plant model consists of a PV array, a boost converter, DC circuit, and a grid-side converter (GSC).

The boost converter controller utilizes the MPPT mode to fetch the maximum power output from the PV panels. A table function in the MPPT determines a voltage reference, V_{MPPT} , at the array based on the line current, I_{PV} , and internal voltage, V_{PV} , of the PV array. The MPPT power reference, P_{ref} , is calculated through an integral controller based on the error between V_{MPPT} and V_{PV} . A limiter has been added after the integrator to limit the P_{ref} within its capacity. The controller then generates switching signals for the boost converter to fetch the maximum power from the array.

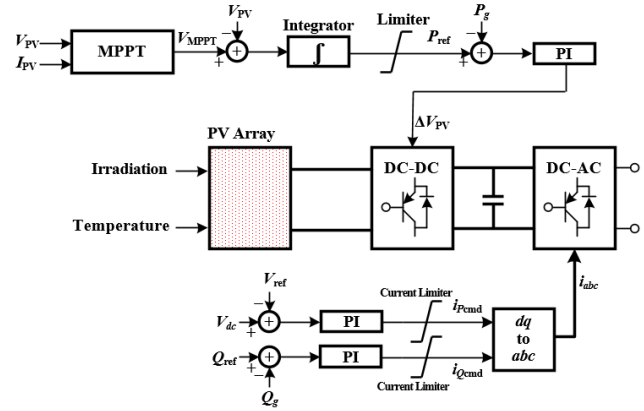


Fig. 1. Configuration of a PV system model.

The GSC, on the other hand, continually manages the DC link voltage, V_{dc} , and the reactive power, Q_g , at the GSC terminal. To achieve the former, the GSC controller generates an active current reference, i_{Pcmd} , using a proportional-integral controller and a current limiter. For the latter, the controller generates a reactive current reference, i_{Qcmd} , through a PI controller and a current limiter. The current limiters have been introduced to limit the currents within the inverter capacity. In this paper, the GSC is simplified as a current source. Thus, the current references are being transformed into three-phase references and fed into the current source, as shown in Fig. 1.

B. Reactive POD scheme in the Inverter Controller

The Reactive POD scheme controls the reactive power output of the PV plant to damp the low-frequency electromechanical oscillations. The speed of the generator has been taken as feedback for the Reactive POD in [25],[29]. The

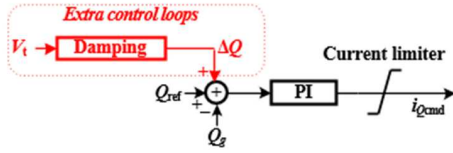


Fig. 2. Block diagram of Reactive POD controller mode of FCS in DC-DC converter.

feedback associated with the speed might have involved delay due to the inertia of these huge synchronous machines. In addition, communication delay referred in [30]–[31] might also significantly impact the performance of the PODs. Hence, instead of using the remote signal from distant areas, we use the bus voltage at the terminal, V_t , of the PV plant as the feedback for this loop. Hence, the reactive power injection during the fault and post fault condition is regulated by the PV plant's terminal voltage.

The signal of V_t is fed to the washout filter to remove the steady state component. The reactive power reference, for the damping control, ΔQ , is obtained by:

$$\Delta Q = V_t \times G \times \frac{sT_w}{1+sT_w} \quad (1)$$

where G and T_w are the gain and the time constant of the washout filter, respectively. T_w is set to 10 sec in this paper.

C. Active POD scheme in the Boost Converter Controller

Fig. 3 shows the typical characteristics curve of 250 kW PV panel simulated in PSCAD.

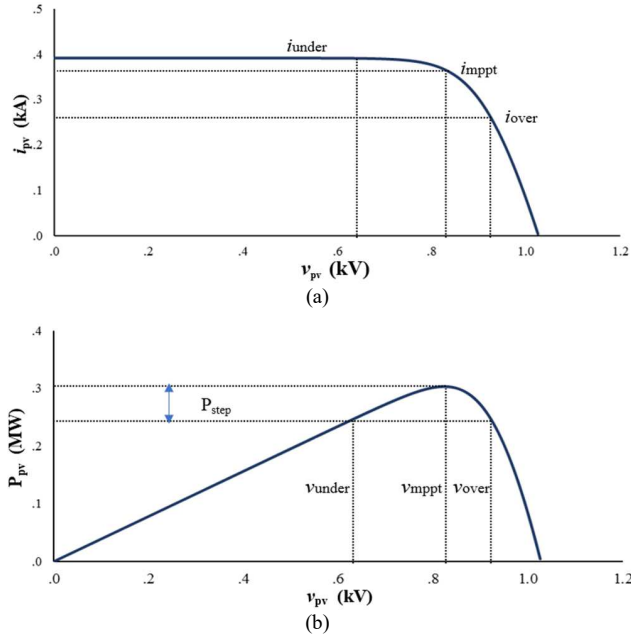


Fig. 3. Typical PV characteristics curves. (a) current, and (b) power versus PV voltage curve.

A PV plant is usually operated on v_{mpp} , to extract the maximum power from the PV arrays to accumulate greater revenues. Nevertheless, any value below MPP can be achieved

by deviating the dc voltage from this optimal operating point. Hence, an extra headroom of active power can be provided by the PV plant if it is operated on under or overvoltage conditions. This residual real power can be utilized in active power damping, an auxiliary service provided by the PV plant. From the characteristic curves, it can be observed that the slope between v_{mpp} and v_{over} is much steeper than that of v_{mpp} and v_{under} . Thus, a more dynamic response is expected if the MPPT is operated on in v_{over} condition. With proper knowledge of dc-link capacitor and future IGBT switches with a slightly higher voltage rating, we can eliminate the harmful effect of overvoltage.

Fig. 4 shows the Active POD scheme used in the boost converter. An additional control function ΔV has been included to modulate the MPPT. The value of ΔV can be found by:

$$\Delta V = K_{dv} \cdot v_{pv} \quad (2)$$

where K_{dv} is a constant and can be determined from the look up table of the characteristics curve.

In pre-fault condition the reference power for the PV panel is $P_{ref} = P_{mpp}$ and $\Delta V = 0$. Once an oscillation is detected, the active POD gets activated. An increase of PV panel voltage within its voltage limits will provide an extra margin ΔP to damp the oscillation. To achieve this oscillation damping, the terminal voltage angle, δ , is taken as the input for the damping control loop. Voltage angle is a key indicator of system stress. It has been observed that voltage angle measurements are very effective in detecting inter-area oscillations. Then, δ is being filtered by a washout filter. The residual real power function feedback for the damping control, ΔP is obtained by a washout filter. ΔP is obtained by:

$$\Delta P = \delta \times G_1 \times \frac{sT_{w1}}{1+sT_{w1}} \quad (3)$$

where G_1 and T_{w1} are the gain and the time constant of the washout filter, respectively. T_{w1} is set to 10 sec in this paper.

When the oscillation is damped effectively for a certain time, the value of ΔV and ΔP gradually again comes to zero.

D. Active and reactive POD controller scheme or Dual POD controller scheme

In this controller, the full capacity of the PV plant is utilized efficiently. The active power capacity of the plant is given priority for active POD in this scheme, while the remaining reactive power capacity is also fully utilized for reactive POD.

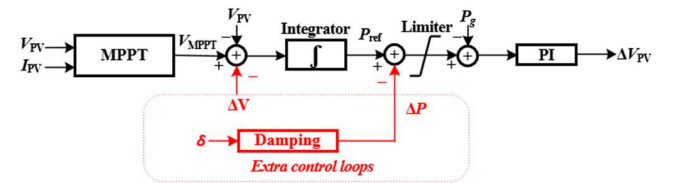


Fig. 4. Block diagram of Active POD controller mode of FCS in DC-DC converter.

III. CASE STUDIES

This paper investigates the efficacy of the PV plants in inter-area low-frequency oscillation in a two area system. For this, a two-area system consisting two 2 GW hydro generators has been developed in PSCAD. A 100 MW large scale PV has been installed in area 2, as shown in Fig. 5. A regular dual POD with the proposed dual POD is made to examine the effectiveness of the damping controller proposed in the paper during the inter-area low-frequency oscillation. There are no PSSs connected to the hydro generators while the PV plant supposes to serve like the PSS.

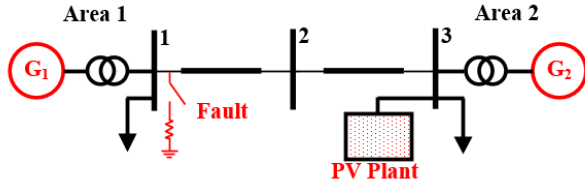
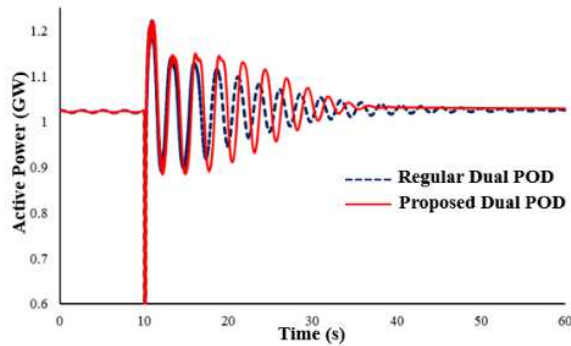


Fig. 5. Two-area system with the PV plant in area 2 and fault in area 1.

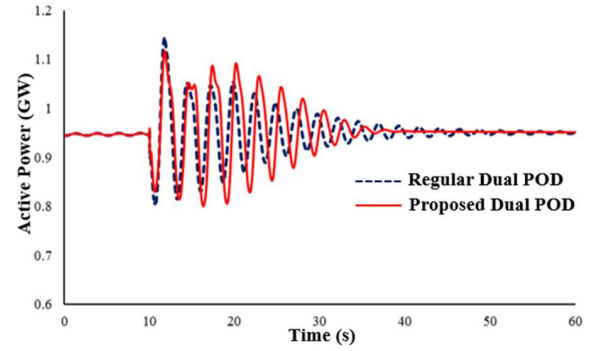
For the regular dual POD the feedback signal is been taken from the speed of the nearest connected synchronous generator. A delay of 0.3s and a sampling of 3Hz are introduced to the signal to replicate the communication delay that might be associated with the use of signals from distant areas.

Each of the dual POD control modes will be observed separately with the same fault in the system. A three-phase to ground fault for 0.2 s is implemented in area 1. We will try to damp out the oscillation from a PV plant connected in area 2.

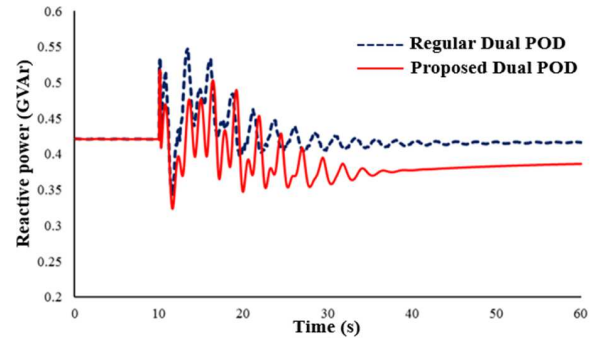
Fig. 6 (a) and (b) depict the active power output of area 1 and 2, respectively. From Fig. 6, we concluded that both the damping controllers work efficiently to damp the inter-area oscillation. However, the regular dual POD is taking more time to damp the oscillation than the proposed dual POD. Thus, the proposed dual POD can damp the oscillation efficiently in much faster time than the other regular controller cited in this paper.



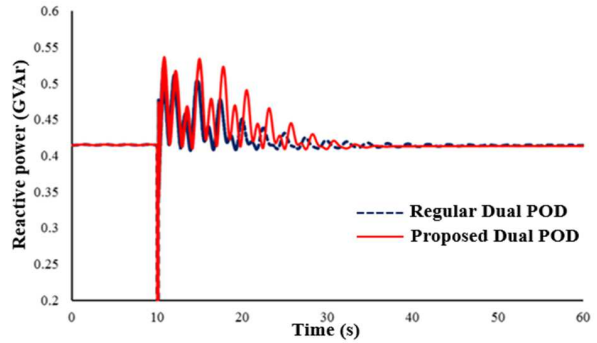
(a)



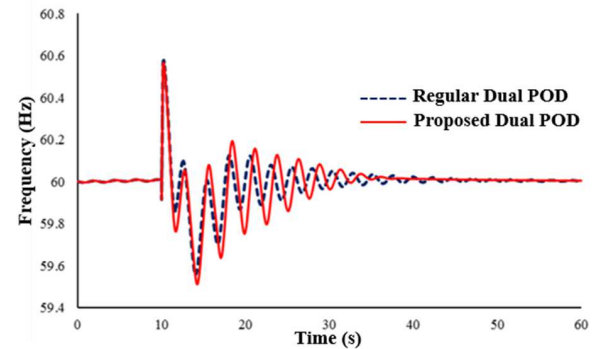
(b)



(c)



(d)



(e)

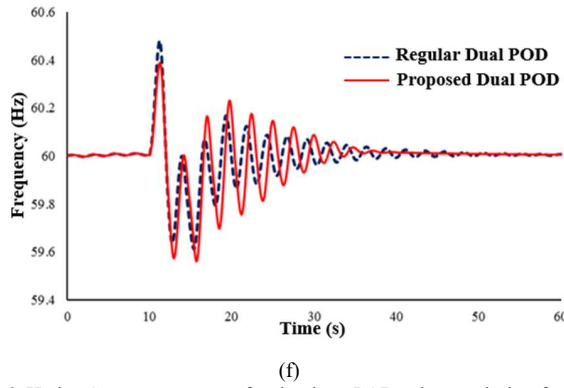


Fig. 6. Hydro Generators output for the three POD schemes during fault and post fault condition. (a) active power output of Generator 1 in area 1. (b) active power output of Generator 2 in area 2. (c) reactive power output of Generator 1. (d) reactive power output of Generator 2 (e) frequency output of Generator 1. (f) frequency output of Generator 2.

Fig. 6 (c) and (d) show the generators' reactive power output plots in areas 1 and 2. Except for the proposed POD curve for generator 1, reactive power output for all the schemes gets to its initial position once the oscillation is over. For generator 1, the reactive power output has not gone to its initial position. This might be because of the PV plant providing extra reactive power for damping the oscillation.

Fig. 6 (e) and (f) depict the frequency at the generator terminals for each area. Alike, the power oscillation, the frequency oscillation is also getting damped efficiently for the proposed POD. The performance of traditional POD schemes also lacks speed in terms of frequency damping. The proposed

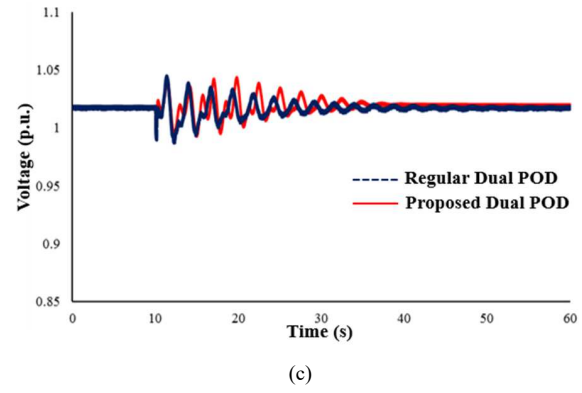
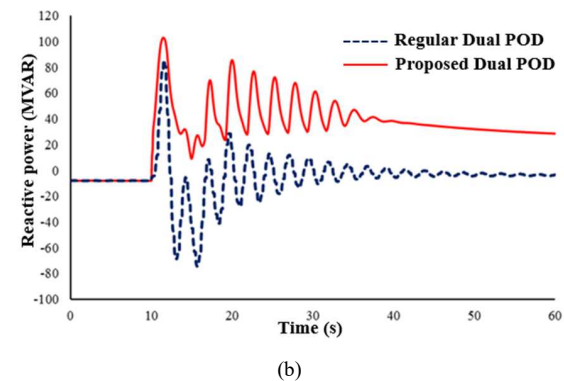
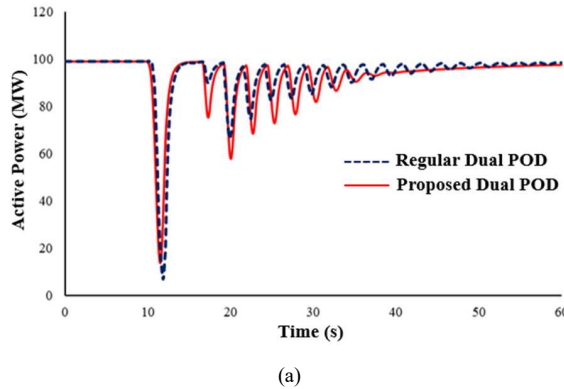


Fig. 7. PV plant output at the POC during fault and post fault condition. (a) active power output. (b) reactive power output. (c) voltage output.

POD is damping out the oscillation within 30 s the fault is applied.

The active power output in Fig. 7 (a), has provided the damping to the system without curtailing its power. The reactive power output in Fig. 7 (b), has also provided voltage stability to the system during fault and post fault conditions. The reactive output of the proposed POD will automatically go to its initial value after there is no detection of fault for some time. The voltage shown in Fig. 7 (c), also remained near to the nominal value 1 p.u. during and after the fault. It also stabilizes once the fault is over.

Utilizing the active and reactive POD using local PMU measurements made the damping extremely fast and effective. Hence, it can be concluded that the proposed dual POD is much more efficient than the other conventional schemes.

IV. CONCLUSION

In this paper, the potential of utilizing the local PMU measurements in PV plant PODs in mitigating inter-area oscillations has been successfully demonstrated. The proposed control scheme is based on adding a damping signal to the active and reactive power control loop of the PV plant. Due to the strong connection to the reactive power injection and terminal bus voltage, the local PMU based voltage measurement is employed for the local signal input of the reactive POD. In the same way, the relation between active power flow and monitoring system stress made the PMU phase angle measurement as the feedback for the active POD. The proposed controller was tested on a two-area system and compared with the conventional regular dual POD. The time-domain simulation proved the effectiveness of the proposed POD over the regular PODs available in the market.

REFERENCES

- [1] The United Nations Framework Convention on Climate Change, "The Kyoto Protocol," [Online] Available: <http://unfccc.int/resource/docs/co/nvkp/kpeng.pdf>, 1997.
- [2] International Renewable Energy Agency, IRENA, "Global Energy Transformation: A roadmap to 2050," Abu Dhabi, Apr. 2018. [Online]

Available: https://www.irena.org/media/Files/IRENA/Agency/Publication/2018/Apr/IRENA_Report_GET_2018.pdf

- [3] E. Kabir, P. Kumar, S. Kumar, and A. A. Adelodun, K. -H. Kim, "Solar energy: potential and future prospects," *Renewable Sustain. Energy Rev.*, vol. 82, pp. 894–900, Feb. 2018.
- [4] Solar Power Europe, "Global market outlook for solar power 2019-2023," Solar Power Europe, Brussels, Belgium, Tech. Rep., May 2019. [Online]. Available: <http://www.solarpowereurope.org/global-market-outlook-2019-2023/>
- [5] U.S. Energy Information Administration, "Annual Energy Outlook 2021," U.S. Department of Energy, Washington DC, US, Tech. Rep., Feb. 2021. [Online]. Available: https://www.eia.gov/pressroom/presentations/AEO2021_Release_Presentation.pdf
- [6] W. Du, H.F. Wang, and L.-Y. Xiao, "Power system small-signal stability as affected by grid-connected photovoltaic generation," *Euro. Trans. Electric. Power*, vol. 21, no. 5, pp. 688–703, July 2011.
- [7] R. Shah, N. Mithulananthan, and KY. Lee, "Contribution of PV systems with ultra-capacitor energy storage on inter-area oscillation," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, pp. 1–8, Jul. 2011.
- [8] R. Shah, N. Mithulananthan, and R.C. Bansal, "Oscillatory stability analysis with high penetrations of large-scale photovoltaic generation," *Energy Convers. Manage.*, vol. 65, pp. 420–429, Jan. 2013.
- [9] R. Doherty, A. Mullane, G. Nolan, D.J. Burke, A. Bryson, and M. O'Malley, "An assessment of the impact of wind generation on system frequency control," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 452–460, Feb. 2010.
- [10] J.M. Carrasco, L.G. Franquelo, J.T. Bialasiewicz, E. Galván, R.C. PortilloGuisado, M.M. Prats, J.I. León, and N. Moreno-Alfonso, "Power-electronic systems for the grid integration of renewable energy sources: A survey," *IEEE Trans. Ind. Electro.*, vol. 53, no. 4, pp. 1002–1016, Jun. 2006.
- [11] Y. Sun, Z. Zhang, G. Li and J. Lin, "Review on frequency control of power systems with wind power penetration," in *International Conf. Power Syst. Tech.*, 2010, pp. 1–8.
- [12] R. Yan, N. -Masood, T. Kumar Saha, F. Bai and H. Gu, "The Anatomy of the 2016 South Australia Blackout: A Catastrophic Event in a High Renewable Network," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 5374–5388, Sept. 2018.
- [13] A. Safayet, P. Fajri, and I. Husain, "Reactive power management for overvoltage prevention at high pv penetration in a low-voltage distribution system," *IEEE Trans. Ind. Appl.*, vol. 53, no. 6, pp. 5786–5794, Nov. 2017.
- [14] Q. A. Al-Shetwi, M. Z. Sujod, and F. Blaabjerg, "Low voltage ride-through capability control for single-stage inverter-based grid-connected photovoltaic power plant," *Solar Energy*, vol. 159, pp. 665–681, Nov. 2018.
- [15] X. Y. Bian, Y. Geng, K. L. Lo, Y. Fu, and Q. B. Zhou, "Coordination of PSSs and SVC Damping Controller to Improve Probabilistic Small-Signal Stability of Power System With Wind Farm Integration," *IEEE Trans. Power Syst.*, vol. 31, pp. 2371–2382, May 2016.
- [16] M. Haque, "Improvement of first swing stability limit by utilizing full benefit of shunt FACTS devices," *IEEE Trans. Power Syst.*, vol. 19, no. 4, pp. 1894–1902, Nov. 2004.
- [17] N. Mithulananthan, C. A. Canizares, J. Reeve, and G. J. Rogers, "Comparison of PSS, SVC, and STATCOM controllers for damping power system oscillations," *IEEE Trans. Power Syst.*, vol. 18, no. 2, pp. 786–792, 2003.
- [18] A. M. Simões, D. C. Savelli, P. C. Pellanda, N. Martins, and P. Apkarian, "Robust design of a TCSC oscillation damping controller in a weak 500-kV interconnection considering multiple power flow scenarios and external disturbances," *IEEE Trans. Power Syst.*, vol. 24, no.1, pp. 226–236, 2009.
- [19] S. Arabi, H. Hamadanizadeh, and B. B. Fardanesh, "Convertible static compensator performance studies on the NY state transmission system," *IEEE Trans. Power Syst.*, vol. 17, no. 3, pp. 701–706, Aug. 2002.
- [20] R. Shah, N. Mithulananthan, and R. C. Bansal, "Damping performance analysis of battery energy storage, ultracapacitor and shunt capacitor with large-scale PV plants," *Appl. Energy Special Issu. Smart Grid*, vol. 96, pp. 235–244, Aug. 2012.
- [21] C. Zhang, X. Cai, Z. Li, A. Rygg, and M. Molinas, "Properties and physical interpretation of the dynamic interactions between voltage source converters and grid: Electrical oscillation and its stability control," *IET Power Electronics*, vol. 10, no. 8, pp. 894–902, Feb. 2017.
- [22] G.B. Huka, W. Li, P. Chao, and S. Peng, "A comprehensive LVRT strategy of two-stage photovoltaic systems under balanced and unbalanced faults," *Int. J. Electric. Power Energy Syst.*, vol.103, pp. 288–301, Dec. 2018.
- [23] R. K. Varma and H. Maleki, "PV solar system control as STATCOM (PV-STATCOM) for power oscillation damping," *IEEE Trans. Sustain. Energy*, pp. 1–1, 2018.
- [24] R. K. Varma and M. Akbari, "Simultaneous Fast Frequency Control and Power Oscillation Damping by Utilizing PV Solar System as PV-STATCOM," *IEEE Trans. Sustain. Energy*, vol. 11, no. 1, pp. 415–425, Jan. 2020.
- [25] M. Basu, V. R. Mahindara, J. Kim, and E. Muljadi, "Effect of high penetrated reactive power support based Inverter-Based-Resources on the power stability of microgrid distribution system during faults," in *Proc. IEEE Ind. Appl. Soc. Ann. Meeting*, 2020, pp. 1–4.
- [26] M. Singh, A. Allen, E. Muljadi and V. Gevorgian, "Oscillation damping: A comparison of wind and photovoltaic power plant capabilities," in *Proc. IEEE Symp. Power Electron. Mach. Wind Water App.*, 2014, pp. 1–7.
- [27] H. D. Silva-Saravia, H. A. Pulgar, L. M. Tolbert, D. A. Schoenwald and W. Ju, "Enabling Utility-Scale Solar PV Plants for Electromechanical Oscillation Damping," *IEEE Trans. Sustain. Energy*, vol. 12, no. 1, pp. 138–147, Jan. 2021.
- [28] I. Abdulrahman, R. Belkacemi, and G. Radman, "Power oscillations damping using wide-area-based solar plant considering adaptive time-delay compensation," *Energy Syst.*, Aug. 2019.
- [29] M. Basu, V. R. Mahindara, J. Kim, R. M. Nelms and E. Muljadi, "Comparison of Active and Reactive Power Oscillation Damping With PV Plants," *IEEE Trans. Ind. Appl.*, vol. , no. , pp., 2021.
- [30] I. Abdulrahman, and G. Radman, "Wide-Area-Based Adaptive Neuro-Fuzzy SVC Controller for Damping Interarea," *Canadian J Electric. Com. Engg.*, vol. 41, no. 3, pp. 133–144, Oct. 2018.
- [31] M. Moradi, and P. Maghouli, "Wide area oscillation damping using utility-scale pv power plants capabilities," *Int. J. Electric. Comp. Engg.*, vol. 7, no. 2, pp. 681–691, Apr. 2017.